

ANALYSTLAB AFRICA

Week 4 Project Summary

HealthConnect Clinic Experience Lab — Data Analytics Track

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1. Track-Specific Problem

As the Data Analytics track representative on the HealthConnect project, my focus is understanding patterns in patient appointment attendance — specifically identifying which factors are associated with the clinic's 48.5% no-show rate, so that later stages can support targeted interventions to reduce missed appointments.

2. Resources Used

HealthConnect_Appointment_Data.csv (5,000 records, 18 fields) and HealthConnect_Data_Dictionary.pdf were both reviewed directly. A full data quality assessment was run on the raw dataset using Python (pandas), and a cleaned version was saved separately without altering the original file.

3. Key Observations from Week 4

- The dataset is high quality: no duplicate records, no logical inconsistencies (e.g. no negative lead times, no appointment dates preceding booking dates), and only minor missing data in three columns
- The overall no-show rate is 48.5% — nearly half of all scheduled appointments are missed, confirming this is a significant operational problem worth investigating further
- Booking lead time shows the clearest early pattern: appointments booked 8+ days in advance have roughly double the no-show rate (51.5%) of same-day or within-week bookings (24.7% / 28.3%)
- Reminders show only a modest effect (47.4% vs. 51.4% no-show rate with vs. without), weaker than expected — worth exploring by reminder channel in Week 5

4. Proposed Approach

Data quality assessment and cleaning are complete. Week 5 will extend the exploratory analysis (e.g. reminder channel breakdown, age and distance relationships) and produce initial visualisations for the five proposed KPIs, ahead of dashboard development.

5. Key Considerations

- The booking-lead-time pattern is the strongest signal found so far and should be validated further (e.g. controlled for appointment type) before being treated as a leading driver
- KPI definitions should stay coordinated with whatever target variable the Data Science track selects, to keep the shared project consistent

6. Proposed Focus for Week 5

Deepen the exploratory analysis on booking lead time and reminder channel, build initial visualisations for the five proposed KPIs, and begin scoping the interactive dashboard structure.